

COMP 9200
Applied Research
Methods

Statistical significance testing

Overview

- Nomenclature
- Correlation coefficient
- Confidence intervals
- Significance testing
- t-tests

Some nomenclature

- A **population** includes each element from the set of observations that can be made.
- A **sample** consists only of observations drawn from the population.
- **Parameter**: measurable characteristic of population (e.g., mean or standard deviation)
- **Statistic**: measurable characteristic of sample

Some nomenclature

- **Percentile**: values that divide a rank-ordered set of elements into 100 equal parts
- **Standard score (z-score)**: how many standard deviations an element is from mean

$$z = (X - \mu) / \sigma$$

z: z-score

X: value of element

μ : mean of population

σ : standard deviation

Some nomenclature

- **Mean**: the average of a set of numbers
- **Median**: The number found in the middle when looking at the set of numbers from smallest to largest
- **Mode**: most commonly occurring value in a set of numbers
- **Variance**: a measure of how data points differ from the mean
- **Standard Deviation**: square root of variance
- **Confidence interval** (more later)

Correlation coefficient

Measure strength of association between 2 variables

- Usually use Pearson product-moment correlation coefficient
- The value of a correlation coefficient ranges between -1 and 1.
- The greater the absolute value of a correlation coefficient, the stronger the linear relationship.
- The strongest linear relationship is indicated by a correlation coefficient of -1 or 1.

Correlation coefficient

Measure strength of association between 2 variables

- The weakest linear relationship is indicated by a correlation coefficient equal to 0.
- A positive correlation means that if one variable gets bigger, the other variable tends to get bigger.
- A negative correlation means that if one variable gets bigger, the other variable tends to get smaller.

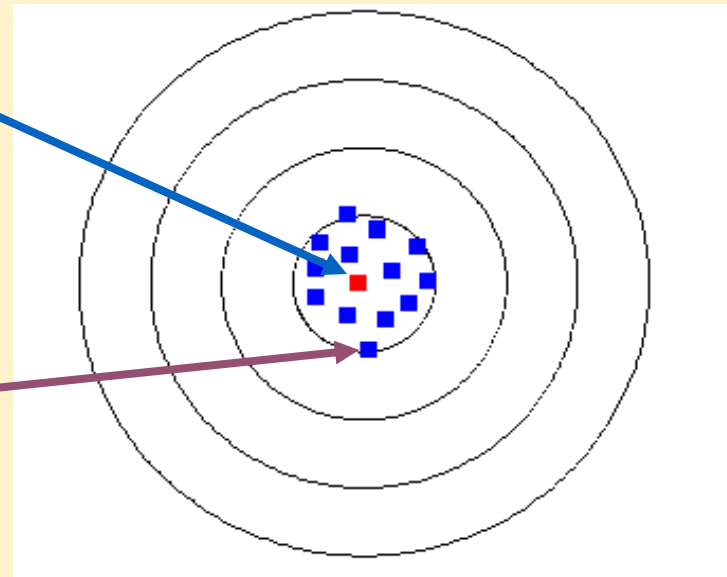
Confidence Intervals

- Statistic estimates population parameter
 - Average salary
 - Mean latency
 - Average FPS
 - Average sales
- Unbiased estimate – on average statistic equals parameter of interest

Confidence Intervals

Unbiased and small variance.

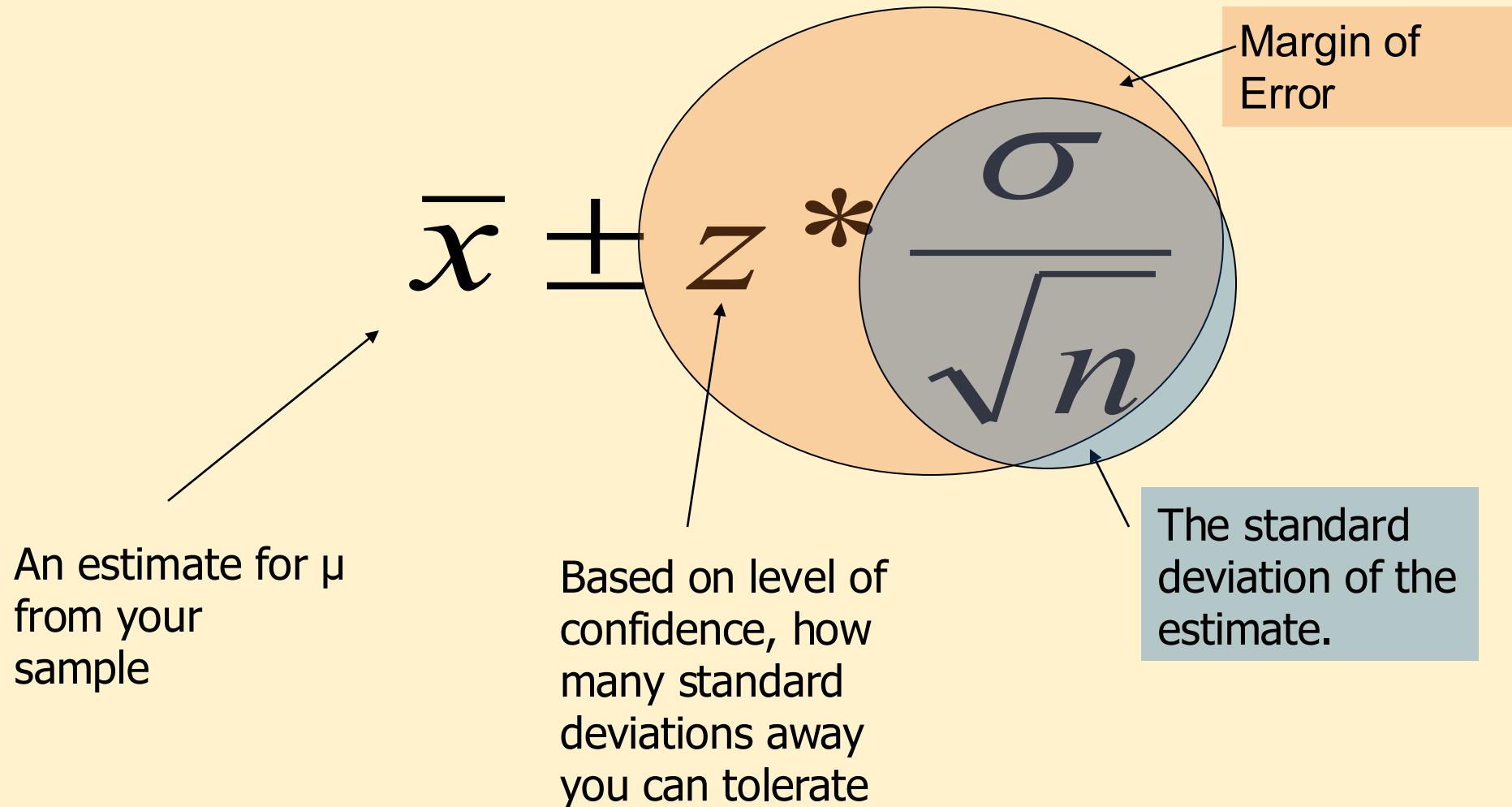
- The center is the target (the parameter of interest).
- The blue points are statistics. Each one is based on a different sample. On average, they equal the target.



Confidence Intervals

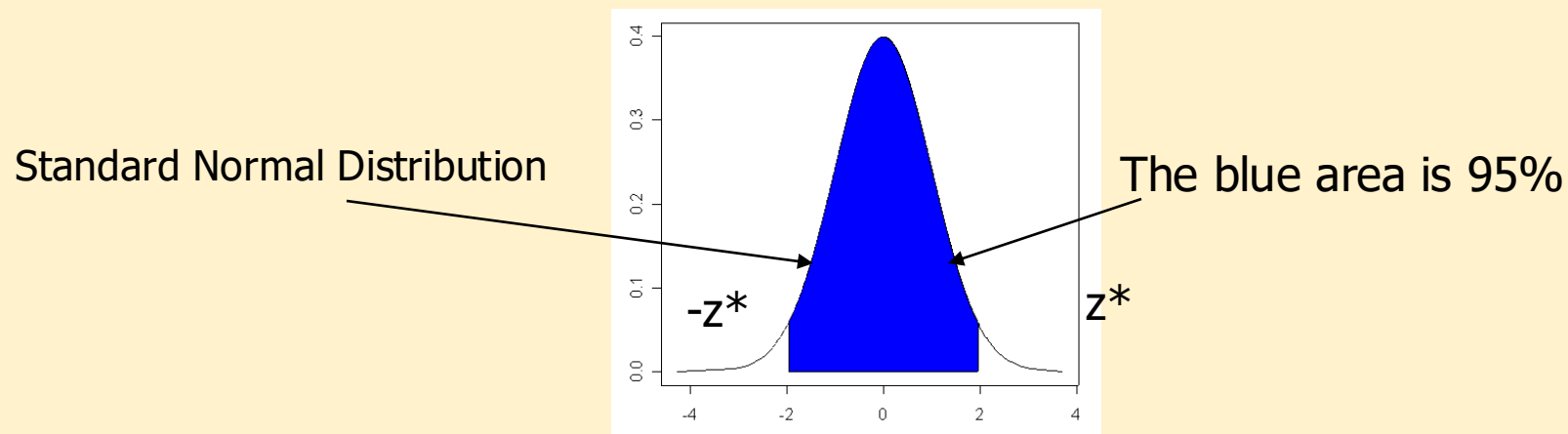
- Confidence Intervals use:
 - the statistic (mean) as a point estimate of the parameter.
 - the standard deviation of the distribution to help provide a sense of accuracy.
 - a measure of confidence in the effectiveness of the interval capturing the true mean (parameter).
 - First find the mean from your sample.
 - Then, find your margin of error based upon your confidence level (z^*) and sample size (n).
 - The population standard deviation, for now, will be assumed to be known.

Confidence Intervals



Confidence Intervals

- z^* (also called the critical value) is the score corresponding to the confidence level you want
- A 95% confidence level means that 95% of intervals of sample size n will capture (or contain) the population mean.
 - Find z^* using the picture and a table or InvNorm on the calculator
- A 99% confidence level means that 99% of intervals of sample size n will capture (or contain) the population mean.



Confidence Intervals

Confidence Level is how confident we want to be that the confidence interval WILL contain the parameter of interest.

Confidence Level	z^*
80%	1.282
90%	1.645
95%	1.96
98%	2.326
99%	2.576

Confidence Intervals

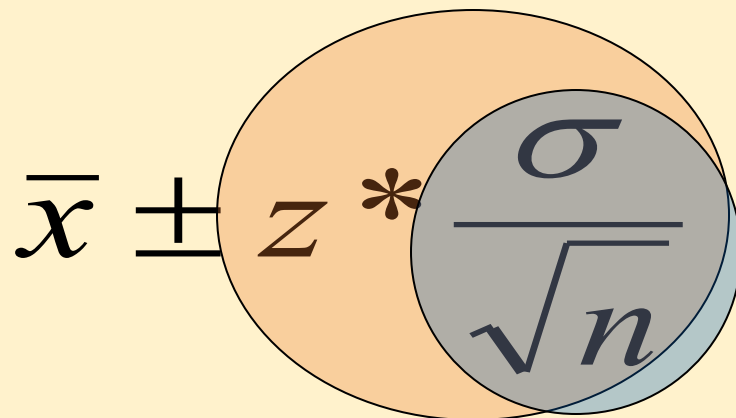
Example #1

- Based on a sample of 40 cars of a particular model, the fuel tank capacity is calculated for each. Based on this data, the sample mean is 18.92 gallons. The population standard deviation is to be 3.5 (remember this is the unrealistic part – we would not really know this).
- Construct a 95% confidence interval for the mean fuel capacity of this model of car.

Confidence Intervals

Example #1

- 95% of all possible confidence intervals for a sample size of 40 will contain the population mean
- We are 95% confident that the mean fuel capacity is between 17.8 and 20.0 gallons.



A diagram illustrating the confidence interval formula. It features two overlapping circles. The left circle is orange and contains the text $\bar{x} \pm$. The right circle is grey and contains the text $z * \frac{\sigma}{\sqrt{n}}$. The overlapping area is shaded blue.

$$\bar{x} \pm z * \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} \pm z * \frac{\sigma}{\sqrt{n}}$$
$$18.92 \pm 1.960 \frac{3.5}{\sqrt{40}}$$
$$18.92 \pm 1.0847$$
$$(17.8353, 20.0047)$$

Confidence Intervals

- We stated that “We are 95% confident that the mean fuel capacity is between 17.8 and 20.0 gallons.”
- Does this mean that $P(17.8 < \mu < 20.0) = 0.95$?
- The answer is NO.
- Suppose that the meteorologist forecasts that there is a 10% chance that it will rain this weekend.
- After Monday comes, does that mean there’s a 10% chance that it rained that past weekend? No, either it rained or it didn’t. So either there is a 0% or a 100% chance that it rained over the past weekend.

Confidence Intervals

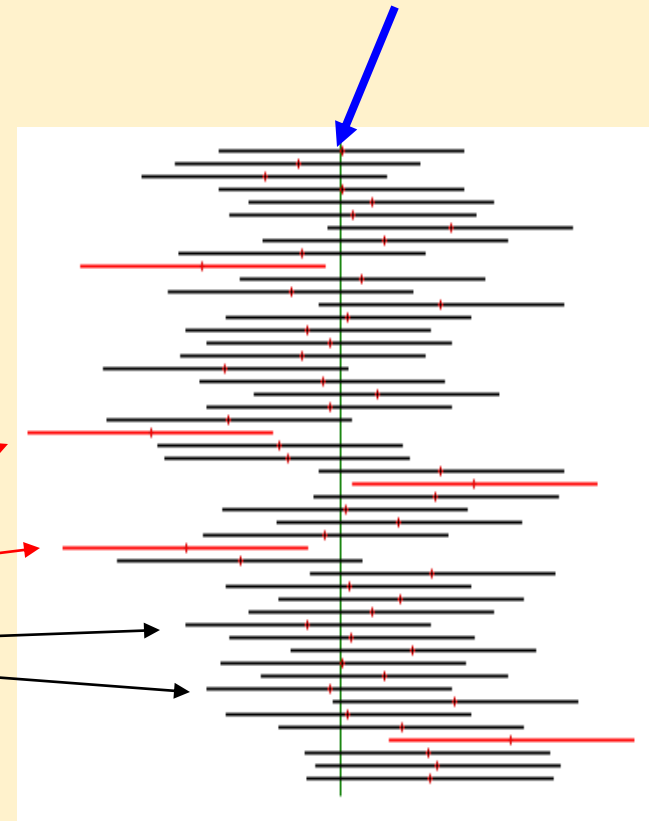
- BEFORE you collect the data, there's a 95% chance that μ will be in the interval but ...
- AFTER you collect the data, either μ is in the interval or it isn't so the probability is either 0 or 1.
- The fact that we don't know what μ is doesn't change this.

Confidence Intervals

50 confidence intervals are obtained. Each one is a 90% confidence interval based on a sample of size n from the same population. We EXPECT approximately 90% of these to contain μ .

The red intervals don't contain μ but the black ones do.

The target (μ). If an interval crosses this vertical line, then μ is in the confidence interval.



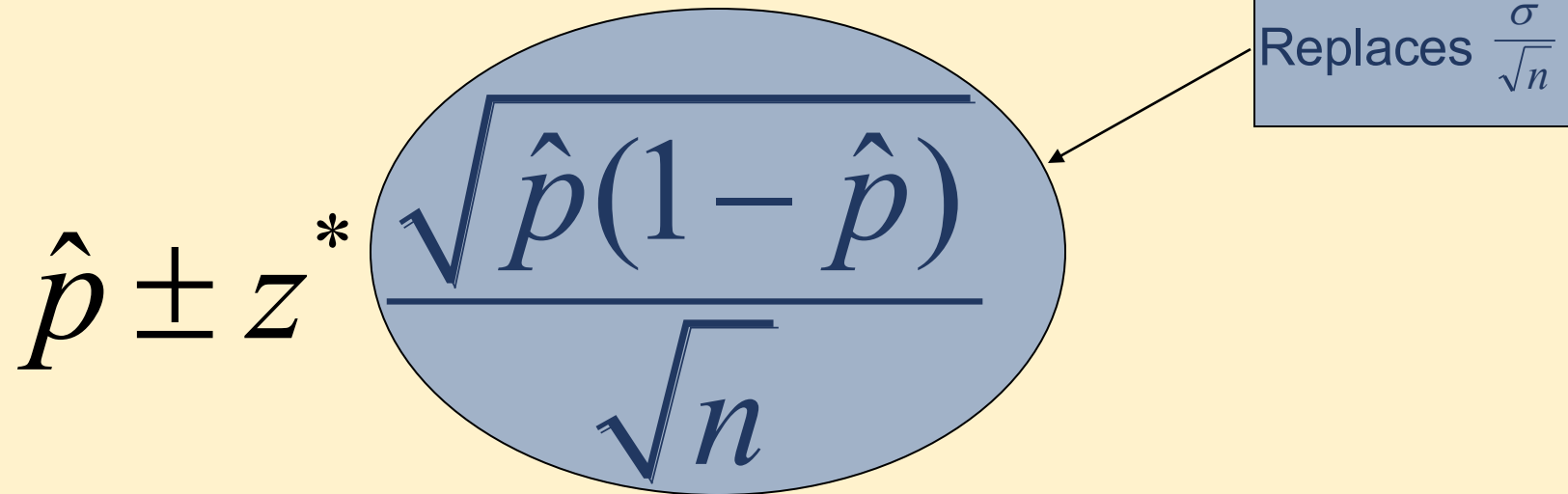
Confidence Intervals

Confidence Intervals for proportions

A confidence interval for p is given by

$$\hat{p} \pm z^* \frac{\sqrt{\hat{p}(1 - \hat{p})}}{\sqrt{n}}$$

Replaces $\frac{\sigma}{\sqrt{n}}$



This is the same formula, just with the standard deviation for proportions in place of the standard deviation for means

Confidence Intervals

Example #2

- Suppose that a sample of 100 light bulbs that a plant produces is chosen at random. In the sample, there are 6 defective bulbs. Obtain a 98% confidence interval for the proportion of defective bulbs they produce.
- Remember that the confidence interval we're using is based on the normal distribution. We should check the rule of thumb to determine if we can use it or not. We can use it since

$$1. n\hat{p} = 100(6/100) = 6 \geq 5$$

$$2. n(1-\hat{p}) = 100(1 - 6/100) = 94 \geq 5$$

Confidence Intervals

Example #2

A 98% confidence interval for the proportion of defective bulbs produced is

$$\begin{aligned} & \hat{p} \pm z^* \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \\ &= 0.06 \pm 2.326 \frac{\sqrt{0.06(1-0.06)}}{\sqrt{100}} \\ &= 0.06 \pm 0.0552 \\ &= (0.0047, 0.1152) \end{aligned}$$

Given the interval (0.0047, 0.1152), we expect somewhere between 0.5% and 11.5% of the bulbs to be defective.

Significance testing

Example #3

- A bottling company claims to put 12 fluid ounces of soda into every can with a standard deviation of $\sigma=0.18$. We suspect they are short changing us. So, we take a sample of 24 cans and measure the mean amount of soda in the cans. We find the mean amount of soda to be 11.92 fluid ounces. Is this enough evidence to suggest that the company is not being truthful?

Significance testing

Example #3

- Assume company innocent until proven guilty
 - Assume there is 12 fluid ounces of soda in each can
 - Take a sample (24 cans) and measure (it came out with a mean of 11.92 fluid ounces)
 - Determine the probability of getting a mean of 11.92 fluid ounces provided that our assumption is true (company is telling the truth)
- This essentially amounts to following the scientific method

Significance testing

- 1) Write hypotheses
 - Null hypothesis (Assume innocence)
 - The mean amount of soda in a can is 12 fluid ounces.
 - Alternate hypothesis (the crime they are charged with)
 - The mean amount of soda in a can is less than 12 fluid ounces
- 2) Calculations
 - Test Statistic
 - A measure of how many standard deviations away from the mean the sample was (a z-score)
 - P-value
 - The probability of the sample mean occurring given the null hypothesis is true
- 4) Conclusions
 - Statistical conclusion
 - Reject the null hypothesis (in favor of the alternate) or Fail to reject the null hypothesis
 - Contextual conclusion
 - State your conclusion in the context of the problem

Significance testing

- The null hypothesis is denoted H_0 .
- This is pronounced “H – not” or “h sub-o”.
- The null hypothesis is a belief about the world, or population.
 - It is typically presuming innocence.
 - It is typically no change or things are as they should be.
 - It is typically what we are trying to show is false.

Significance testing

- The alternative hypothesis is denoted H_a .
- This hypothesis is pronounced "H-sub a" or alternative hypothesis or alternate hypothesis.
 - This is typically your (or the study's) suspicion.
 - This is typically a belief about the population that you believe to be true.
- There are only 3 possible alternative hypotheses
 - $H_a : \mu < \mu_0$ The mean is less than the null hypothesis
 - $H_a : \mu > \mu_0$ The mean is greater than the null hypothesis
 - $H_a : \mu \neq \mu_0$ The mean is different than the null hypothesis

Significance testing

- The test statistic is a convenient summary of the sample data that can be easily used to make decisions about the hypotheses.
- It is often some kind of transform of the data to a more convenient form, like Z-scores.
- Test statistic is calculated under assumption the H_0 is true.
- It can be thought of as how many standard deviations away from the mean the sample was.

Significance testing

- A p-value is a probability, so it is a number that is between 0 and 1.
- It is a measure of how consistent the sample data is with the null hypothesis.
- There are two definitions of p-value that will be useful to you:
 - The p-value is the probability of observing a value of the test statistic as extreme or more extreme than the one observed, if the H_0 is true.
 - The p-value is the probability of observing data like ours if the null hypothesis is true.

Significance testing

- Most of scientific world uses .05 as a standard p-value or cut-off point.
- P-values less than .05 are considered small chances and those above .05 are large.
- This gets ridiculous when p-values = .0501 or .049999. P-values are in shades of gray, not black and white.

Significance testing

- The conclusion gives our final opinion about what the data tells us about our hypotheses.
 - Was the evidence of the data enough to convince us that the null hypothesis is not plausible?
 - Or...was there not enough evidence to refute the null hypothesis?

Significance testing

Example #3

- For the null hypothesis, assume the company is innocent and really does put 12 fluid ounces in each can.
 - $H_0 : \mu = 12$
- For the alternative hypothesis, we think they are not giving us enough soda, so...
 - $H_a : \mu < 12$

Significance testing

Example #3

- For the calculations, we use the sampling distribution based upon the null hypothesis and a sample size of 24 cans.
 - Assuming normality (normal population, sample size and Central Limit Theorem), the sampling distribution should be N:

$$N\left(12, \frac{0.18}{\sqrt{24}}\right)$$

$$N(12, 0.0367)$$

Significance testing

Example #3

- Our sample had a mean of 11.92 fluid ounces.
 - Find the test statistic (remember a z-score).

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{11.92 - 12}{0.0367}$$

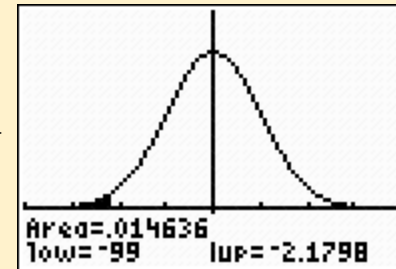
$$z = -2.1798$$

Significance testing

Example #3

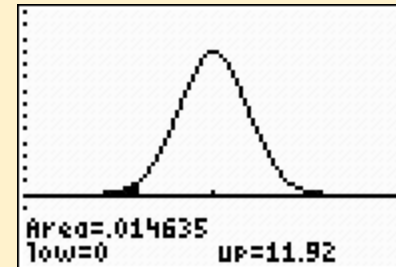
- Our sample had a mean of 11.92 fluid ounces.
 - Find the P-value (the probability of this happening given the null hypothesis is true).

$$p = \text{normalcdf}(-99, -2.1798) = 0.0146 \longrightarrow$$



or

$$p = \text{normalcdf}(0, 11.92, 12, .0367) = 0.0146 \longrightarrow$$



Significance testing

Example #3

- Interpret the P-value
 - There is only a 1.46% chance of us getting the sample we did given that company was telling the truth.
- Generally speaking, if $p < .05$ or $p < 5\%$, it is considered enough evidence to reject the null hypothesis.
- Since $.0146 < .05$, we have enough evidence!
- Statistical conclusion
 - Reject the null hypothesis in favor of the alternative hypothesis
- Contextual conclusion
 - There is sufficient evidence to suggest that company is putting less than 12 fluid ounces of soda into their cans.

Significance testing

Example #4

- Another bottling company claims claims that one of their energy drinks has 40 mg of sodium per can. We work for a consumer organization that tests such claims. We take a random sample of 50 cans and find that the mean amount of sodium in the sample is 41.4 mg. The standard deviation in all cans is 7.2 mg. We suspect that there is more than 40 mg of sodium per can.

Significance testing

Example #4

- For the null hypothesis, assume the company is innocent and the soda really does contain 40 mg of sodium in each can.
 - $H_0 : \mu = 40$
- For the alternative hypothesis, we think there may be more than 40 mg of sodium, so...
 - $H_a : \mu > 40$

Significance testing

Example #4

- For the calculations, we use the sampling distribution based upon the null hypothesis and a sample size of 50 cans.
 - Assuming normality (normal population, sample size and Central Limit Theorem), the sampling distribution should be N:

$$N\left(40, \frac{7.2}{\sqrt{50}}\right)$$

$$N(40, 1.0182)$$

Significance testing

Example #4

- Our sample had a mean of 42.6 mg of sodium.
 - Find the test statistic (remember a z-score).

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{41.4 - 40}{1.0182}$$

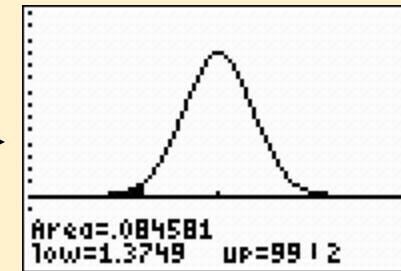
$$z = 1.3749$$

Significance testing

Example #4

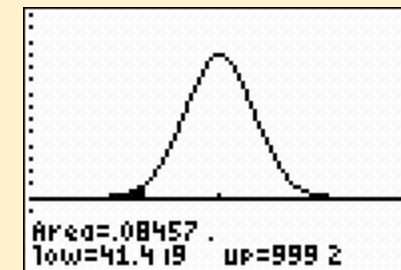
- Our sample had a mean of 42.6 mg of sodium.
 - Find the P-value (the probability of this happening given the null hypothesis is true).

$$p = \text{normalcdf}(1.3749, 99) = 0.0846 \longrightarrow$$



or

$$p = \text{normalcdf}(41.4, 999, 40, 1.0182) = 0.0846 \longrightarrow$$



Significance testing

Example #4

- Interpret the P-value
 - There is an 8.46% chance of us getting the sample we did given that the company was telling the truth.
- Generally speaking, if $p < .05$ or $p < 5\%$, it is considered enough evidence to reject the null hypothesis.
- Since $.0846 > .05$, we do not have enough evidence!
- Statistical conclusion
 - Fail to reject the null hypothesis in favor of the alternative hypothesis
- Contextual conclusion
 - There is not sufficient evidence to suggest that the energy drink has more than 40 mg of sodium in a can of soda.

Significance testing

Example #5

- Let's suppose that a computer shop typically sell 3400 items of your company's products per week (with a standard deviation of 500). You then begin a new marketing campaign complete with a new slogan and mass marketing strategy. You are curious how the new advertising will affect sales (to see if sales increased or decreased). You sample 40 random stores for a week and find that the mean number of items the stores sell is 3550.

Significance testing

Example #5

- For the null hypothesis, assume sales do not change and the stores still sell 3400 items per week.
 - $H_0 : \mu = 3400$
- For the alternative hypothesis, we want to see if there is a change (for better or worse), so...
 - $H_a : \mu \neq 3400$

Significance testing

Example #5

- For the calculations, we use the sampling distribution based upon the null hypothesis and a sample size of 40 stores.
 - Assuming normality (normal population, sample size and Central Limit Theorem), the sampling distribution should be N:

$$N\left(3400, \frac{500}{\sqrt{40}}\right)$$

$$N(3400, 79.0569)$$

Significance testing

Example #5

- Our sample had a mean of 3550 items.
 - Find the test statistic (remember a z-score).

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{3550 - 3400}{79.0569}$$

$$z = 1.8974$$

Significance testing

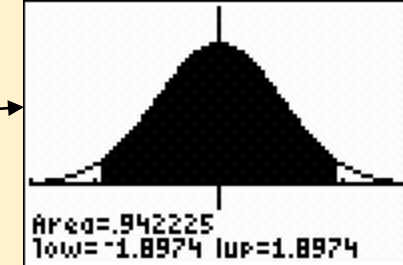
Example #5

- Our sample had a mean of 3550 items.
 - Find the P-value (the probability of this happening given the null hypothesis is true).

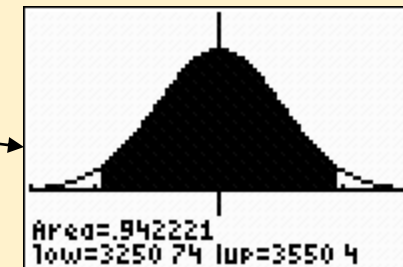
$$p = 2\text{normalcdf}(1.8974, 99) = 0.0578$$

or

$$p = 2\text{normalcdf}(3550, 99999, 3400, 79.0569) = 0.0578$$



It is the un-shaded area that we want



Significance testing

Example #5

- Interpret the P-value
 - There is an 5.78% chance of us getting the sample we did given that the stores typically sell 3400 items per week.
- Generally speaking, if $p < .05$ or $p < 5\%$, it is considered enough evidence to reject the null hypothesis.
- Since $.0578 > .05$, we do not have enough evidence!
- Statistical conclusion
 - Fail to reject the null hypothesis in favor of the alternative hypothesis
- Contextual conclusion
 - There is not sufficient evidence to suggest that the marketing campaign had an impact on the sales of your products.

t-tests

When you know the standard deviation of the population use a Z-Test.

The test statistic is z .

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Use the normal distribution to calculate your p-value.

When you **don't** know the standard deviation of the population use a **T-Test**.

The test statistic is **t** .

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

Use a **t-distribution with $(n-1)$ degrees of freedom** to calculate your p-value.

t-tests

Checking Assumptions

- CAUTION!
- The t-test is NOT always appropriate!
- In order to appropriately use the t-test:
 - The population distribution must be normal
 - OR
 - The sample size must be large enough

t-tests

How large a sample is large enough?

Sample Size	Check
$n < 15$	In order to use t-procedures, the sample distribution should be pretty normal. There should not be any outliers and very little skewness.
$15 < n < 40$	In order to use t-procedures, there should not be any large outliers and there should not be extreme skewness.
$n > 40$	The sample size is large enough to use t-procedures regardless.

t-tests

Example #6

- A manufacturer of a special bolt requires that this type of bolt have a mean shearing strength in excess of 110 lb. To determine if the manufacturer's bolts meet the required standards a sample of 25 bolts was obtained and tested. The sample mean was 112.7 lb and the sample standard deviation was 9.62 lb. Use this information to perform an appropriate hypothesis test with a significance level of 0.05.

t-tests

Example #6

μ = the mean shearing strength of this specific type of bolt

The hypotheses to be tested are

$$H_0: \mu = 110 \text{ lb}$$

$$H_a: \mu > 110 \text{ lb}$$

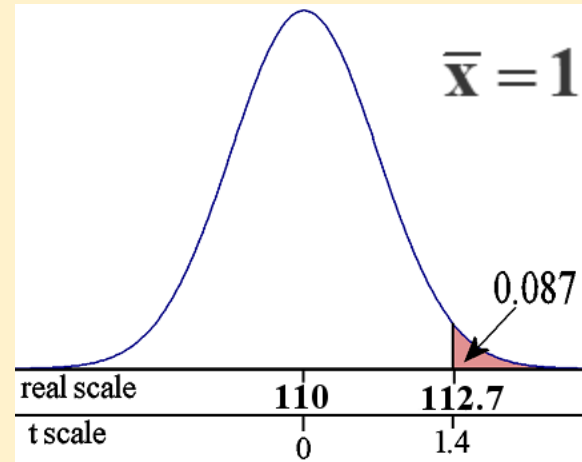
The significance level to be used for the test is $\alpha = 0.05$.

The test statistic is

$$t = \frac{\bar{x} - 110}{s / \sqrt{n}}$$

t-tests

Example #6



$$\bar{x} = 112.7, s = 9.62, n = 25, df = 24$$

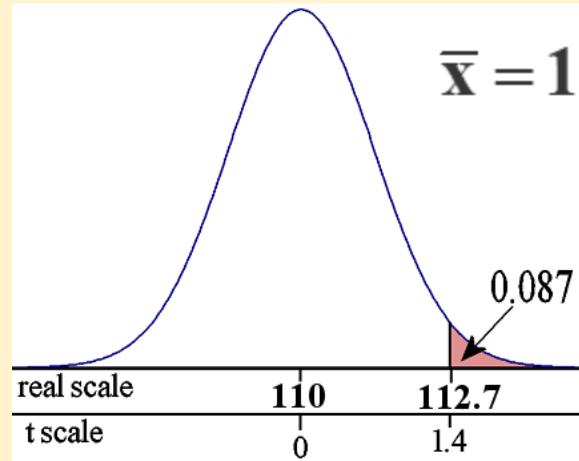
$$\text{P-value} = P\left(t > \frac{112.7 - 110}{9.62 / \sqrt{25}}\right)$$

$$= P(t > 1.4) = 0.087$$

t Table											
cum. prob	t .50	t .75	t .80	t .85	t .90	t .95	t .975	t .99	t .995	t .999	t .9995
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745

t-tests

Example #6



$$\bar{x} = 112.7, s = 9.62, n = 25, df = 24$$

$$\begin{aligned} \text{P-value} &= P\left(t > \frac{112.7 - 110}{9.62 / \sqrt{25}}\right) \\ &= P(t > 1.4) = 0.087 \end{aligned}$$

Because $p\text{-value} = 0.087 > 0.05 = \alpha$, we fail to reject H_0 .

At a level of significance of 0.05, there is insufficient evidence to conclude that the mean shearing strength of this brand of bolt exceeds 110 lbs.

t-tests

Example #6

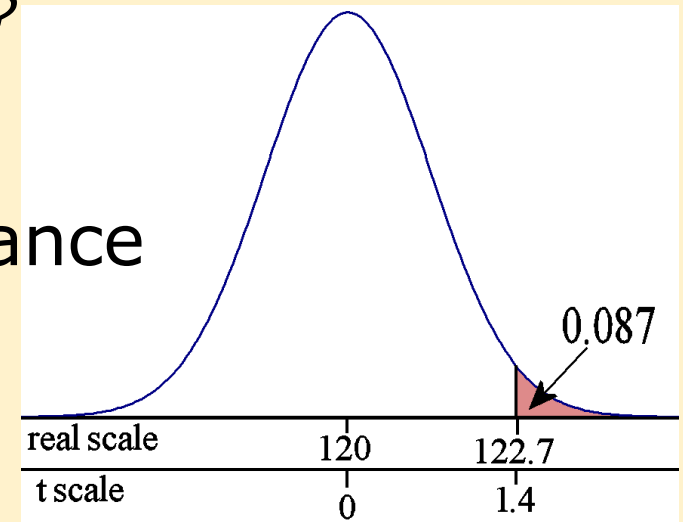
What if significance level was 0.10 instead of 0.05?

$$p\text{-value} = 0.087 < 0.10 (\alpha)$$

We can reject H_0 at 0.10 level of significance

Now we can conclude:

At the 0.10 level of significance there is sufficient evidence to conclude that the mean shearing strength of this brand of bolt exceeds 120 lbs.



t-tests

- Many people are bothered by the fact that different choices of α lead to different conclusions.
- This is natural of a process where you control the probability of being wrong.
- Remember, when you select the level of significance, you choose your willingness to accept a certain chance of a Type I error.

t-tests

Example #7

A jeweler is planning on manufacturing gold charms. His design calls for a particular piece to contain 0.08 ounces of gold. The jeweler would like to know if the pieces that he makes contain (on the average) 0.08 ounces of gold. To test to see if the pieces contain 0.08 ounces of gold, he made a sample of 16 of these particular pieces and obtained the following data.

0.0773	0.0779	0.0756	0.0792	0.0777
0.0713	0.0818	0.0802	0.0802	0.0785
0.0764	0.0786	0.0776	0.0793	0.0755
0.0806				

Use a level of significance of 0.01 to perform an appropriate hypothesis test.

t-tests

Example #7

1. The population characteristic being studied is μ = true mean gold content for this particular type of charm.
2. Null hypothesis: $H_0: \mu = 0.08$ oz
3. Alternate hypothesis: $H_a: \mu \neq 0.08$ oz
4. Significance level: $\alpha = 0.01$
5. Test statistic:

$$t = \frac{\bar{x} - \text{hypothesized mean}}{s / \sqrt{n}} = \frac{\bar{x} - 0.08}{s / \sqrt{n}}$$

t-tests

Example #7

Computations:

$$n = 16, \bar{x} = 0.077891, s = 0.0025143$$

$$t = \frac{0.077891 - 0.08}{0.0025143 / \sqrt{16}} = -3.2$$

p-value: Two tailed test!

Look up table for t curves, $t = 3.2$, $df = 15$ we get 0.003

$$p\text{-value} = 2(0.003) = 0.006$$

t Table											
cum. prob	t .50	t .75	t .80	t .85	t .90	t .95	t .975	t .99	t .995	t .999	t .9995
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073

t-tests

Example #7

- Conclusion:
 - Since $p\text{-value} = 0.006 \leq 0.01$, we reject H_0
 - At the 0.01 level of significance there is sufficient evidence that the true mean gold content of this type of charm is not 0.08 ounces.
 - Actually when rejecting a null hypothesis for the $\neq$ alternative, a one tailed claim is supported. In this case, at the 0.01 level of significance, there is convincing evidence that the true mean gold content of this type of charm is less than 0.08 ounces.

t-tests

Example #8: Matched Pairs t-Test

A weight reduction center advertises that participants in its program lose an average of at least 5 pounds during the first week of the participation. Because of numerous complaints, the state's consumer protection agency doubts this claim. To test the claim at the 0.05 level of significance, 12 participants were randomly selected. Their initial weights and their weights after 1 week in the program appear on the next slide. Set up and perform an appropriate hypothesis test.

t-tests

Member	Initial Weight	One Week Weight
1	195	195
2	153	151
3	174	170
4	125	123
5	149	144
6	152	149
7	135	131
8	143	147
9	139	138
10	198	192
11	215	211
12	153	152

t-tests

Member	Initial Weight	One Week Weight	Difference Initial -1week
1	195	195	0
2	153	151	2
3	174	170	4
4	125	123	2
5	149	144	5
6	152	149	3
7	135	131	4
8	143	147	-4
9	139	138	1
10	198	192	6
11	215	211	4
12	153	152	1

This last column is the data we are after!

t-tests

Example #8:

μ_d = mean of the individual weight changes (initial weight–weight after one week)

This is equivalent to the difference of means:

$$\mu_d = \mu_1 - \mu_2 = \mu_{\text{initial weight}} - \mu_{\text{1 week weight}}$$

$$H_0: \mu_d = 5$$

$$H_a: \mu_d < 5$$

Significance level: $\alpha = 0.05$

$$\text{Test statistic: } t = \frac{\bar{x}_d - \text{hypothesized value}}{s_d / \sqrt{n}} = \frac{\bar{x}_d - 5}{s_d / \sqrt{n}}$$

t-tests

Example #8:

Calculations: $n = 12, \bar{x}_d = 2.333, s_d = 2.674$

$$t = \frac{\bar{x}_d - 5}{s_d / \sqrt{n}} = \frac{2.333 - 5}{2.674 / \sqrt{12}} = -3.45$$

p-value = 0.0027

t Table											
cum. prob	t .50	t .75	t .80	t .85	t .90	t .95	t .975	t .99	t .995	t .999	t .9995
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437

Conclusions: $p\text{-value} = 0.0027 < 0.05$ so we reject H_0

There is strong evidence that the mean weight loss for those who took the program for one week is less than 5 pounds.

Review

- Nomenclature
- Correlation coefficient
- Confidence intervals
- Significance testing
- t-tests